

4(a)(i)	any three from: I heterochromatin / euchromatin envelope / two membranes / double membrane / inner and outer membrane ; nuclear pores ; A pores in nuclear envelope <i>for two marks</i> (outer surface of) outer membrane with ribosomes ; <i>ref. to</i> (outer nuclear) <u>membrane</u> continuous with RER ; R inner membrane AVP ; e.g. perinuclear space A intermembranous space pore complexes	3
4(a)(ii)	any two from: resolution / resolving power, is, low / lower / poor / AW / 200 nm / 0.2 µm ; (A range 100–300 nm) A electron microscope has a higher resolution further detail ; e.g. ribosomes / ER, smaller than 200 nm can only see cell structures greater than (limit of) resolution cannot see structures smaller than 200 nm cell structures too small for the resolution (<i>needs ref. to resolution</i>) cell structures, too small to / do not, interfere with light waves ; AW	2
4(b)	if letter used in more than one row, R these rows correct interphase knowledge G2 = A, S = B, G1= blank ; <i>cytokinesis</i> blank ; <i>prophase</i> C ; <i>anaphase</i> E ; <i>telophase</i> D ;	5
4(c)	any three from: <i>large numbers of B-lymphocytes / plasma cells (in primary immune response)</i> 1 large quantity of (specific) antibody, produced / released or (large quantity of) antibody to form antibody-antigen complexes / to bind antigen (for phagocytosis) / AW ; <i>large numbers of memory B-lymphocytes so</i> 2 provide long term <u>immunity</u> / memory cells long-lived / provides immunological memory ; A remain in, circulation (for a long time) AW 3 able to produce fast(er), secondary (immune) response ; A second response will be fast(er) A immune response faster on second encounter (with antigen / pathogen) / AW 4 higher concentration / faster production, of antibodies (than primary response) ; I 'more' alone 5 person does not have, symptoms / become ill (of / from, same disease) ; A presence of same, pathogen / antigen, does not cause disease 6 AVP ; memory cells can (divide to) produce plasma cells more plasma cells present than primary response able to form more memory cells	3

Question	Answer	Marks
4(d)	any two from: immune response / antibodies produced, against, self antigens ; I immune system attacks self A autoimmunity / autoimmune disease <i>idea that faulty B-lymphocytes not destroyed ;</i> A <i>ref. to T-lymphocytes if in correct context</i> (specific) antibody, binds to / acts on / AW, self-antigen / receptor, on the (cell surface membranes of) muscle cells / at neuromuscular junction ; A antibody binds to acetylcholine receptors <i>ref. to consequence to muscle cells ; e.g.(nerve) impulse conduction impaired</i> <i>action of transmitter substance hindered</i>	2

Question	Answer	Marks
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